3. Water Jug Problem

Aim: To write a Python program to obtain exactly 2 litres of water using a 4-litre jug and a 3-litre jug.

Algorithm:

1. Initialize both jugs as empty.
2. Fill the 3-litre jug.
3. Pour water into the 4-litre jug.
4. Refill the 3-litre jug when it becomes empty.
5. Empty the 4-litre jug when it becomes full.
6. Repeat until exactly 2 litres are obtained in the 4-litre jug.
7. Display the amount of water in both jugs.

Code:

```
a = 0
b = 0
while a! = 2:
    if b == 0:
        b = 3
    elif a == 4:
        a = 0
    else:
        water = min (b, 4 - a)
        a += water
        b -= water
    print ("4-litre jug:", a)
    print ("3-litre jug:", b)
    print ()
print ("2 litres obtained!")
```

Input:

4-litre jug capacity = 4

3-litre jug capacity = 3

Required water = 2 litres

Output:

```
--------------------
4-litre jug: 0
3-litre jug: 3

4-litre jug: 3
3-litre jug: 0

4-litre jug: 3
3-litre jug: 3

4-litre jug: 4
3-litre jug: 2

4-litre jug: 0
3-litre jug: 2

4-litre jug: 2
3-litre jug: 0

2 litres obtained!
```

Result:

Thus, the Water Jug Problem was solved successfully using Python, obtaining exactly 2 litres of water in the 4-litre jug.